

Ushtrime ND11c

2.7.1 Me çfarë shtypje mund të shtypet $m=10g$ helium në temperaturën $T=273K$ në enën e qelqit me vëllim $V=5l$ (Masa molare e heliumit $M=4 \cdot 10^{-4} kg/mol$).

Zgjidhje:

Të dhënat:

$$m=10g$$

$$T=273K$$

$$V=5l$$

$$M=4 \cdot 10^{-4} \frac{kg}{mol}$$

$$P=?$$

$$PV=nRT$$

$$n=\frac{m}{M}$$

$$P=\frac{nRT}{M \cdot V}$$

$$P=\frac{10 \cdot 10^{-3} kg \cdot 8,314 \frac{J}{mol \cdot K} \cdot 273K}{4 \cdot 10^{-4} \frac{kg}{mol} \cdot 5 \cdot 10^{-3} m^3}$$

$$P=\frac{22697,22 \cdot 10^{-3}}{20 \cdot 10^{-7}} Pa = 1134,861 \cdot 10^4 Pa$$

$$P=113,48 \cdot 10^5 Pa$$

$$P=113,48 bar$$

$$P=\frac{1}{3} n_0 m \langle v^2 \rangle$$

$$n_0=\frac{N}{V}$$

$$E_k=\frac{m \langle v^2 \rangle}{2} \text{ ose } m \langle v^2 \rangle = 2E_k$$

$$P=\frac{1}{3} \frac{N}{V} \cdot E_k$$

$$E_k=E_{\text{med}}=\frac{3}{2} kT=\frac{3}{2} \frac{R}{N_A} \cdot T$$

$$P=\frac{2}{3} \cdot \frac{N}{V} \cdot \frac{3}{2} \frac{R}{N_A} T$$

$$P=\frac{N}{N_A} \cdot \frac{RT}{V} \Rightarrow P=\frac{nRT}{V}$$

$$\text{ose } PV=nRT$$

$$\frac{J}{m^3} = \frac{N \cdot \frac{J}{mol} \cdot \frac{1}{mol}}{m^3} = \frac{N}{m^3} = Pa$$

2.7.2 Sa molekula oksigjen gjenden në enë me vëllim $V=0,5l$ në shtypje $p=980,67mba$ dhe temperaturë $T=300K$?

Zgjidhje:

Të dhënat:

$$V=0,5l=0,5 \cdot 10^{-3} m^3=5 \cdot 10^{-4} m^3$$

$$p=980,67 mbar=980,67 \cdot 10^{-3} \cdot 10^5 Pa=980,67 \cdot 10^2 Pa=9,8067 \cdot 10^4 Pa$$

$$T=300K$$

$$N=?$$

$$N_A=6,022 \cdot 10^{23} \frac{1}{mol}$$

$$R=8,314 \frac{J}{mol \cdot K}$$

$$\frac{Pa \cdot m^3}{J} mol = \frac{\frac{N \cdot \frac{J}{mol}}{mol} \cdot m^3}{N \cdot \frac{J}{mol}} \cdot mol$$

$$PV=nRT \Rightarrow n=\frac{PV}{RT}=\frac{9,8067 \cdot 10^4 Pa \cdot 5 \cdot 10^{-4} m^3}{8,314 \frac{J}{mol \cdot K} \cdot 300K}=\frac{49,0335}{2494,2} mol=0,0196 mol$$

$$n=\frac{N}{N_A} \Rightarrow N=n \cdot N_A$$

$$n=0,0196 mol$$
$$n=1,96 \cdot 10^{-2} mol$$

$$N=n \cdot N_A=1,96 \cdot 10^{-2} mol \cdot 6,022 \cdot 10^{23} \frac{1}{mol}$$

$$N=11,8 \cdot 10^{21} \text{ molekula}$$